

03

A student investigated the effects of indoleacetic acid (IAA) on the growth of oat seedlings (young plants).

The student:

- removed the shoot tip from each seedling and cut out a 10 mm length of shoot
- placed 10 lengths of shoot into each of 5 Petri dishes
- added to each Petri dish an identical volume of 5% glucose solution
- added to each Petri dish 40 cm³ of a different concentration of IAA solution
- left the Petri dishes at 20 °C in the dark with their lids on for 5 days
- removed the shoots after 5 days and measured them
- determined the mean change in length of shoot at each concentration of IAA.

Table 1 shows her results.

Table 1

IAA concentration added to Petri dish / parts per million	10 ⁻⁵	10 ⁻³	10 ⁻¹	1	10
Mean change in length of shoot / mm	0.0	0.1	1.3	2.4	3.1

03

1

Explain why the student removed the shoot tip from each seedling.

[2 marks]

[Extra space]

0 3 . 2

Explain why the student added glucose solution to each Petri dish.

[2 marks]

0 3 . 3

Explain why the lids were kept on the Petri dishes.

[2 marks]

0 3 . 4

Describe and explain the results shown in **Table 1** and suggest how the results might have differed if lengths of **root** had been used.

[3 marks]

0 3 . 5

The student produced the different concentrations of IAA using a stock 1 g dm^{-3} solution of IAA ($1\text{ g dm}^{-3} = 1\text{ part per thousand}$) and distilled water.

Complete **Table 2** with the volumes of stock IAA solution and distilled water required to produce 40 cm^3 of 10 ppm (parts per million) IAA solution.

[1 mark]

Table 2

Concentration of IAA solution / parts per million	Volume of stock IAA solution / cm^3	Volume of distilled water / cm^3
10		

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